

Vesper Node Cleaner

Documentation · v1.0.0 · Unreal Engine 5.5 – 5.8

Installation

1. Download the plugin from your FAB Library.
2. Copy the `VesperNodeCleaner` folder into your project's `Plugins/` directory (create that folder at your project root if it doesn't exist yet).
3. Re-launch the project. Unreal will prompt to rebuild the plugin module — accept it.
4. Open **Edit** → **Plugins**, search "Vesper", and confirm **Vesper Node Cleaner** is enabled.

NOTE This is an editor-only plugin. It adds no runtime dependencies and has zero effect on packaged builds.

Quick start

1. Open any Blueprint's Event Graph.
2. Marquee-select two or more nodes.
3. Press `Shift + Q`, or right-click and choose **Vesper: Auto-Format Selected**.

The selected nodes rearrange into columns based on how they're wired together. The change is a single Undo step.

Auto-Format Graph

Arranges the selected nodes into left-to-right columns ("layers") based on execution and data pin connectivity, keeping each node's original top-to-bottom order within its column to minimize wire crossings.

How to run it

METHOD	WHERE
Keyboard shortcut	<code>Shift + Q</code> , while a Blueprint graph has focus
Right-click menu	<i>Vesper: Auto-Format Selected</i> , in the graph's context menu
Toolbar button	<i>Clean Graph</i> , in the Blueprint Editor toolbar

Requirements

- At least 2 nodes must be selected. With fewer, Vesper shows a notification and does nothing else.
- Only connections between two *selected* nodes are considered — links to nodes outside the selection are ignored, so you can safely format a small piece of a large graph.

Auto-Fit Comment Box

Resizes the selected Comment box so it tightly wraps the nodes currently inside it, with consistent padding on all sides.

How to run it

METHOD	WHERE
Keyboard shortcut	Shift + C , while a Blueprint graph has focus
Right-click menu	<i>Vesper: Auto-Fit Comment Box</i> , in the graph's context menu
Toolbar button	<i>Fit Comment</i> , in the Blueprint Editor toolbar

TIPRun *Auto-Format Graph* first, then draw or select your Comment box, then run *Auto-Fit Comment Box* — fitting after formatting avoids resizing twice.

Keyboard shortcuts

ACTION	DEFAULT SHORTCUT
Auto-Format Graph	Shift + Q
Auto-Fit Comment Box	Shift + C

Both can be remapped under **Edit** → **Editor Preferences** → **Keyboard Shortcuts**, search "Vesper".

How the layout works

Vesper treats your selection as a small directed graph, not a row of shapes to snap into place. Three passes:

1. Read connections

For every selected node, Vesper looks at its input pins and finds which other *selected* nodes feed into it.

2. Compute layers

A node with no selected predecessors is layer 0. Every other node's layer is one more than the deepest layer among its predecessors (longest-path layering). This repeats until layers stabilize.

3. Place nodes

Each layer becomes a column, spaced a fixed distance apart. Within a column, nodes keep their original top-to-bottom order — this cheap heuristic meaningfully reduces wire crossings without a full crossing-minimization pass.

WHY THIS DIFFERS FROM NATIVE ALIGN Unreal's built-in Align

Top/Bottom/Middle and Align Left/Center/Right snap the current selection to one shared axis. They don't read pin connections, so they can't tell a downstream node from an upstream one, or separate two branches of a Branch node. Vesper's layering is connection-aware by design.

Known limitations

- **Node size is estimated, not measured.** Unreal doesn't store a node's rendered pixel size in the graph data — only Comment boxes do. Vesper estimates height from pin count, which is generous enough to avoid overlap on all standard K2 node types but isn't pixel-perfect.
- **Auto-Reroute is not included in v1.0.0.** It's on the roadmap for a future update.
- **Cyclic selections** (two nodes that feed each other, e.g. through an Event/Delay loop) are layered using a bounded relaxation pass — they'll still lay out sensibly, but layering a true cycle is inherently a best-effort approximation.

Troubleshooting

Nothing happens when I click Auto-Format

- Confirm at least 2 nodes are selected — with fewer, Vesper shows a notification instead of an error.
- Confirm a Blueprint Graph Editor tab is actually focused (click into the graph once, then try again).

Shift + Q doesn't do anything

KNOWN BEHAVIOR The shortcut only fires when focus is on a widget that has the Vesper command list bound. Clicking the toolbar button or using the right-click menu always works; if the shortcut specifically doesn't, use one of those instead while this is refined in an upcoming update.

Toolbar icons look like placeholders

Cosmetic only — icon artwork is versioned independently from plugin logic and is updated in point releases without changing how the commands behave.

Still stuck?

Email support@alwaysprettystore.com with your Unreal Engine version and, if possible, a screenshot of the graph before running the command.

Changelog

VERSION	NOTES
1.0.0	Initial release — Auto-Format Graph, Auto-Fit Comment Box.
